

Practice Midterm

Form P · Practice

Practice for the midterm exam · not graded · use it to time yourself

Name: _____ Student ID: _____

40 points | Suggested working time 40 minutes

Use a basic calculator. Show your working. Allow about 21 minutes for Part 1, 7 for Part 2, 6 for Part 3 and 3 for the Summary, plus 3 minutes to read and check. Part 1 and the Summary are marked on clear reasoning and clear explanations in your own words. The data are synthetic teaching data from a fictional seaside café. This practice trains the same skills as the real midterm on different data; in Part 2 you practise the standard deviation, which the real midterm builds into the t-test.

Part	What it covers	Points	Score
1	Concepts: model choices, interpreting coefficients, outliers	24	
2	Calculations by hand: standard deviation	5	
3	Computations in Sheets: functions and interpretation	8	
Summary	Summary for the owner	3	
Total		40	

PART 1 · CONCEPTS: model choices and interpretation

24 points · about 21 minutes

Q1 · Choose the inputs (7 points, about 6 minutes)

A seaside café wants to predict **tomorrow's pastry sales** (pastries) **before it opens**. Table 1 lists information it could use. Across 168 days of data, **the planned price and the new-price-period flag are identical ($r = 1.00$)**.

Table 1. Candidate information.

#	Information
1	Tomorrow's temperature forecast (°C)
2	Tomorrow's wind forecast (km/h)
3	Whether tomorrow is a weekend (yes or no)
4	Whether a beach event is on the calendar tomorrow (yes or no)
5	Tomorrow's planned pastry price (€)
6	A "new price period" flag: 0 before 30 June, 1 from 30 June (the day the price rose)
7	Tomorrow's gelato sales (cups), counted at closing time

1a Choose **three** inputs for the model. For each, say whether it should push pastry sales **up or down**, and give a one-line reason. 3 pts

Input (name or number)	Up or down?	Reason (one line)

1b One candidate in Table 1 is **unsuitable** for this forecast. Which one, and what is the problem? 2 pts

1c Should the planned price **and** the new-price-period flag both go into the same model? Explain, and say which one you would keep. 2 pts

Q2 · Interpret the coefficients (14 points, about 12 minutes)

A model of daily pastry sales (pastries) was fitted to 168 days with $\text{=LINEST}(y, X, \text{TRUE}, \text{TRUE})$. $X_1 = \text{temperature } (^\circ\text{C})$, $X_2 = \text{wind (km/h)}$, $X_3 = \text{weekend (1 = Saturday or Sunday, 0 = weekday)}$. In the data, temperature ran from 13 to 32 °C and wind from 3 to 26 km/h. Table 2 shows how LINEST lays out its results; Table 3 is the output.

Table 2. How LINEST lays out its results.

Col A (X_3)	Col B (X_2)	Col C (X_1)	Col D (Intercept)	Statistic description
Slope m_3	Slope m_2	Slope m_1	Intercept (b)	Coefficients for: $y = m_1X_1 + m_2X_2 + m_3X_3 + b$
SE – m_3	SE – m_2	SE – m_1	SE – Intercept	Standard error for each coefficient
R^2	SE – Y	Blank	Blank	Quality of the model fit
F-statistic	df	Blank	Blank	Analysis of variance metrics
SS – Reg	SS – Resid	Blank	Blank	Sum of squares metrics

Table 3. LINEST output (shaded cells are blank).

Col A	Col B	Col C	Col D
11.53	–0.18	–0.81	84.76
1.53	0.10	0.20	4.98
0.33	8.95		
26.34	164		
6329.40	13133.74		

2a Write the fitted equation, using the numbers in Table 3 and the variable names.

2 pts

2b (i) Interpret the **temperature** coefficient: what happens to predicted pastry sales when temperature rises by 1 °C, and what is held constant? (ii) In your own words, why do we say “holding the other variables constant”?

3 pts

2c (i) The **temperature** coefficient is negative. What does the sign mean, and what is one possible real-world reason? (ii) The **weekend** coefficient is positive and weekend is a 0/1 variable. Interpret it in one sentence. (iii) A 1 km/h change in wind is tiny (wind ranges from 3 to 26 km/h). Rewrite the wind coefficient for a **10 km/h** increase.

3 pts

2d (i) Rewrite the temperature coefficient for a **10 °C** increase. (ii) Which is the better message for the owner, “per 1 °C” or “per 10 °C”? Why? (iii) A colleague says: “The weekend coefficient (11.53) is much bigger than the wind coefficient (–0.18), so weekends matter far more than wind.” Is that a fair comparison? Use your answer to 2c(iii).

3 pts

2e Tomorrow is forecast to be 20 °C with a 12 km/h wind on a weekend. (i) Predict pastry sales. (ii) The wind turns out to be **10 km/h stronger** than forecast. By how much would predicted sales change, everything else the same?

3 pts

Q3 · Outliers (3 points, about 3 minutes)

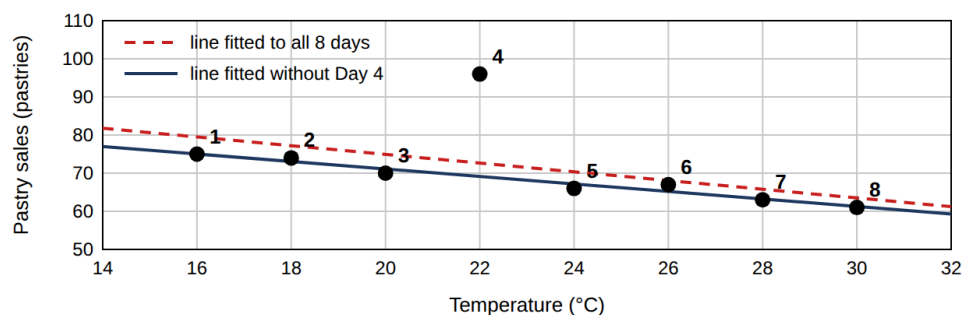
Constructed example: pastry sales on 8 days (Table 4, Figure 1) and the fitted line with and without Day 4 (Table 5).

Table 4. Temperature and sales on 8 days.

Day	1	2	3	4	5	6	7	8
Temperature (°C)	16	18	20	22	24	26	28	30
Sales (pastries)	75	74	70	96	66	67	63	61

Table 5. Results for the straight line (sales against temperature).

	All 8 days	Without Day 4
Slope (pastries per °C)	-1.14	-0.98
R ²	0.26	0.96
SE – Y (pastries)	10.28	1.17

Figure 1. The 8 days, with the line fitted to all of them (dashed) and the line fitted without Day 4 (solid).

3a Which day does not follow the pattern of the others?

1 pt

3b What impact does that day have on the slope, and what impact on R²?

1 pt

3c Before deleting it, what would you do?

1 pt

PART 2 · CALCULATIONS BY HAND: standard deviation

5 points · about 7 minutes

Q4. On 30 June the café raised the pastry price from €2.50 to €2.75. Table 6 shows pastry sales (pastries) on 5 weekdays **before** (Group 1) and 5 **after** (Group 2). Both samples are weekdays with similar weather. Formula as given in class:

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

4a Complete Table 6: the two means, the two sums of squared deviations, and s for each group (2 decimals).

3 pts

Table 6. Worksheet (fill in the blank cells).

Day	Before (x ₁)	x ₁ – $\bar{x}_1$	(x ₁ – $\bar{x}_1$) ²	After (x ₂)	x ₂ – $\bar{x}_2$	(x ₂ – $\bar{x}_2$) ²
1	66			61		
2	70			69		
3	62			57		
4	68			67		
5	64			61		
Sum						
Mean $\bar{x}$						
s						

4b (i) Which group's daily sales are more spread out? (ii) The T.TEST in Part 3 compares the two group means. Why would larger standard deviations make a difference between the means harder to detect?

2 pts

PART 3 · COMPUTATIONS IN SHEETS: functions and interpretation

8 points · about 6 minutes

Q5. On the sheet “Price test”, the “before” sales are in D2:D6 and the “after” sales in E2:E6 (Part 2). On the sheet “Daily data”, pastry sales are in G2:G169 and the three X columns in H2:J169 (Part 1).

- 5a** Write the Sheets formula for each (any correct form is accepted): (i) the sample standard deviation of the “before” sales; (ii) the p-value comparing “before” with “after”, two-tailed, equal variances; (iii) the regression with all its statistics (the LINEST output). 3 pts

- 5b** In `=T.TEST(D2:D6, E2:E6, 2, 2)`, what do the last two numbers mean? In `=LINEST(..., TRUE, TRUE)`, what does the second TRUE add to the output? 2 pts

Sheets gives `=T.TEST(D2:D6, E2:E6, 2, 2) = 0.283`.

- 5c** (i) State the null hypothesis in words. (ii) Is the difference statistically significant at 5%? Explain using the p-value. (iii) Does this result prove the price rise had **no** effect? Explain. 3 pts

SUMMARY FOR THE OWNER

3 points · about 3 minutes

- 6** Write 3–4 sentences to the café owner, who has no statistics background. Include: (i) one input from the Part 1 model and what a sensible change in it does to pastry sales, holding the others constant; (ii) what the price test showed; (iii) one caution. 3 pts

— End of practice —